

Product Laws of Exponents

Teacher's Edition

Read This First

This lesson rebuilds Math Nation 3.1 (SE pp. 99–105) as front-taught notes and individual boards. The book's version is four partner Explorations in a row; every table in them survives here, worked from the front, because a single student can complete each one.

The mathematics in the book's lesson is clean — every value on its pages checks. The defect is in its practice key: the Additional Practice answer for $(-3 \cdot 6)^4$ is printed 104,796 and is 104,976. That item is on the Additional bank here with the right value.

Two things the state guide expects that the book never asks: working backwards to an unknown exponent ($2^3 \cdot 2^n = 2^8$), and the sum-inside-the-parentheses trap $((10 + (-2))^3)$. Both are in the round and in the bank.

Standard and Targets

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Clarification Instruction builds the laws from specific examples; Appendix E lists them. Whole-number exponents only in this lesson — the zero exponent arrives in 3.02 and negative exponents in 3.04.

Boundary Rational-number bases, including negative and decimal bases. No variables until the Thread A days after 3.07.

Learning target I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

Essential question How can the laws of exponents be used to write an equivalent expression?

Building on MA.6.NSO.3.3 — evaluating positive rational numbers with natural-number exponents (Grade 6). Unit 2 — squares and cubes as repeated multiplication.

Working toward MA.8.NSO.1.3 (integer exponents, Lessons 3.04–3.07) and MA.8.AR.1.1 (the same laws with variable bases, the Thread A days after 3.07).

Where this sits in the course First of three lessons carried from MA.7.NSO.1.1 (3.01–3.03). Everything here is rehearsed again with integer exponents under MA.8.NSO.1.3, which the Grade 8 FAST reports under Number Sense and Operations.

Lesson at a Glance

Windy Hill periods run **53 minutes**. The whiteboard round takes the remainder after the taught blocks — **16 minutes** in this lesson — and IXL takes the last 5. Slide numbers below are read from the deck when this edition is built.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-7	14 min
Example 1	8-9	3 min
Your Turn	10-11	3 min
Example 2	12-13	3 min
Your Turn	14-15	3 min
Whiteboards	16-33	16 min
IXL	34	5 min
Total		53 min

Materials Whiteboards and markers. Calculators are allowed on everything and are useful for the big values in the bank (8^8 , 35^4); the laws themselves are done by hand.

Where the lesson lives Notes II (the expansion that proves the product law) and Whiteboard questions 5 and 6 (the two diagnostics that separate 'add' from 'multiply'). If the round shows a board full of B on question 5, the expansion is what to go back to.

No partner or group work anywhere in this lesson. Every Math Nation exploration this lesson draws on was written for pairs; here the same tables are worked from the front and on individual boards.

Vocabulary

base — the number that is used as a factor: in 2^3 the base is 2

exponent — the number of times the base is used as a factor: $2^3 = 2 \cdot 2 \cdot 2$

power — a base with an exponent, such as 2^3 ; its value is 8

product of powers law — $a^m \cdot a^n = a^{m+n}$ — same base, add the exponents

power of a product law — $(ab)^m = a^m \cdot b^m$ — the exponent goes to every factor

power of a power law — $(a^m)^n = a^{mn}$ — multiply the exponents

identity exponent law — $a^1 = a$

Warm-Up — spaced retrieval

Four questions, two minutes silent, then the reveal. No reteach longer than one sentence per question.

1. $\sqrt{81}$ 9

retrieves: yesterday — Unit 2 — the principal square root (2.02)

2. Between which two whole numbers is $\sqrt{40}$? **6 and 7**

retrieves: last week — Unit 2 — approximating roots (2.04): $36 < 40 < 49$

3. Solve $3x - 7 = 11$ **$x = 6$**

retrieves: last unit — Unit 1 — two-step equations (1.04)

4. Evaluate $\left(\frac{2}{3}\right)^2$ **$\frac{4}{9}$**

retrieves: prior grade — the prerequisite this lesson needs — MA.6.NSO.3.3 — a fraction raised to a whole-number power

Slide-by-Slide Teaching Notes

Slide 1. Product Laws of Exponents 1 min

- Post the learning target. Say the 'today' line and nothing else yet.

Slide 2. Warm-Up — On your own. Four minutes. No notes. 4 min

- Q4 is the prerequisite: a rational base is squared by squaring numerator and denominator. A board showing $4/3$ or $2/9$ tells you the base was not treated as one number.

Slide 3. Warm-Up — Answers. 1 min

- Reveal. Ask for the band each question came from only if time allows.

Slide 4. Notes — Copy the three words. 3 min

- Students copy the three vocabulary rows. Say them aloud: base, exponent, power. The exponent COUNTS factors; it is not a factor itself — that is the misconception behind $2^3 = 6$.
- **OFF THE SLIDE, YOURS TO SAY:** Ask: what does 2 to the 3 mean? Insist on the answer 'three 2s multiplied', not 'six'.

Slide 5. Notes — Where the rule comes from. 4 min

- Expand both powers, count the factors, then write the rule. The law only applies when the bases match: $3^2 \cdot 2^3$ has no shortcut, and the reveal on the second row is the reason.
- **OFF THE SLIDE, YOURS TO SAY:** 'We are not adding the bases and not multiplying the exponents. We are counting factors.'

Slide 6. Notes 3 min

- Expand $(2 \cdot 5)^3$ as three copies of $(2 \cdot 5)$, regroup with the commutative and associative properties, and read off $2^3 \cdot 5^3$. The trap is a SUM inside the parentheses: $(10 + (-2))^3$ is 8^3 , and the exponent does not distribute over addition. Question 9 of the round tests exactly this.

Slide 7. Notes 4 min

- $(4^2)^3$ is three copies of 4^2 , so six 4s: multiply the exponents. Then the identity law: one factor of the base is the base. Students who write $4^2 \cdot 3 = 12$ or 4^5 (adding) are confusing this with the product law; the reveal on the table makes the count visible.

- **OFF THE SLIDE, YOURS TO SAY:** 'Product of powers: add. Power of a power: multiply. Both come from counting factors, so count if you forget.'

Slide 8. Example 1 — Two ways to the same value. *1 min*

- SPOKEN ONLY. Read the expression, ask which law applies and why (same base). Do not write yet.

Slide 9. Example 1 — Worked both ways. *2 min*

- Left column: the law, one exponent, then the value. Right column: evaluate each power and multiply. Same 243 either way — that is the check students can always run.

Slide 10. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches a student who multiplied the exponents ($2^{12} = 4096$) or multiplied the bases (4^7). A board reading 128 or 2^7 is right.

Slide 11. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 12. Example 2 — A negative base and a power of a power. *1 min*

- SPOKEN ONLY. The brackets are not absolute value bars — say so before anyone asks. Ask: how many factors of (-2) are there in total?

Slide 13. Example 2 — Worked. *2 min*

- Multiply the exponents: six factors of -2 . An even number of negative factors gives a positive product; the sign question is the one students carry into Lesson 3.02.

Slide 14. Your Turn — Same steps, your numbers. Boards up when done. *2 min*

- Catches the added-exponent error ($(-3)^5 = -243$, a negative answer) and the dropped sign (a student who writes 3^6 gets 729 too — ask them to write the base with its sign).

Slide 15. Your Turn — Answer. *1 min*

- Reveal; name what a wrong board most likely did (see the note above).

Slide 16. Whiteboards · Question 1 of 9 *16 min for the round*

- Product of powers, the plain case. A board with 25^5 multiplied the bases; 5^6 multiplied the exponents.

Slide 17. Whiteboards · Question 1 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 18. Whiteboards · Question 2 of 9

- Same law, now evaluated. $2^8 = 256$ on a board means the exponents were multiplied; 4^6 means the bases were.

Slide 19. Whiteboards · Question 2 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 20. Whiteboards · Question 3 of 9

- Power of a product. Either route lands on 1000. A board with 30 multiplied 10 by 3.

Slide 21. Whiteboards · Question 3 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 22. Whiteboards · Question 4 of 9

- Power of a power. 3^6 means the exponents were added.

Slide 23. Whiteboards · Question 4 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 24. Whiteboards · Question 5 of 9

- Diagnostic. B is the most common wrong board; a board full of B means the two laws are being told apart by the LOOK of the expression rather than by counting factors — go back to the Notes II expansion.

Slide 25. Whiteboards · Question 5 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 26. Whiteboards · Question 6 of 9

- Diagnostic, mirror of Q5. D is rare but instructive: the student treated the exponent as a number to be squared.

Slide 27. Whiteboards · Question 6 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 28. Whiteboards · Question 7 of 9

- Same base with a sign. A board with 27 dropped the sign; a board with -3^3 read as -27 is fine but ask for the parentheses.

Slide 29. Whiteboards · Question 7 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 30. Whiteboards · Question 8 of 9

- Rational base. This is the B1G-M example item for 7.NSO.1.1. A board with $1/10$ multiplied the bases' denominators by the count; $1/6$ multiplied exponents and used $\frac{1}{2}$ once.

Slide 31. Whiteboards · Question 8 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 32. Whiteboards · Question 9 of 9

- Written. The law needs a product; a sum is evaluated first. Accept any answer that says 'add first' and lands on 512. A student who writes $10^3 + (-2)^3 = 992$ has

distributed the exponent over addition — the misconception this question exists to surface.

Slide 33. Whiteboards · Question 9 of 9

- Reveal. Scan the back row first; question the blank boards before the wrong ones.

Slide 34. IXL — Last five minutes. 5 min

- IXL: whatever is not finished is tonight's practice; SmartScore 67.

The Whiteboard Round — what each board tells you

Nine questions, one at a time, each with its reveal. Boards down until the cue, all up together, scan the back row first, question the blank boards before the wrong ones. Question 9 is written work. Every wrong option on a multiple-choice question is a named error; the benchmark it comes from is cited.

#	Question	Answer	What a wrong board says
1	$5^2 \cdot 5^3$	5^5	25^5 — multiplied the bases [7.NSO.1.1 misconception: operating on the bases]; 5^6 — multiplied the exponents [confused with power of a power]
2	$2^4 \cdot 2^2$	64	256 — multiplied the exponents (2^8); 4096 — multiplied the bases (4^6)
3	$(2 \cdot 5)^3$	1000	30 — multiplied the base by the exponent [misconception: exponent treated as a factor]; 250 — applied the exponent to one factor only
4	$(3^2)^4$	3^8	3^6 — added the exponents [confused with product of powers]; 9^8 — squared the base as well
5	Which expression is equivalent?	A — 4^5	(B) multiplied the exponents — the power-of-a-power move applied to a product [7.NSO.1.1 misconception, B1G-M]; (C) multiplied the bases as well as adding the exponents [7.NSO.1.1 misconception: operating on the bases]; (D) added the bases and added the exponents [7.NSO.1.1 misconception: operating on the bases]

#	Question	Answer	What a wrong board says
6	Which expression is equivalent?	A — 2^6	(B) added the exponents — the product-of-powers move applied to a power of a power [7.NSO.1.1 misconception, B1G-M]; (C) raised the base to the outer exponent as well [7.NSO.1.1 misconception: operating on the bases]; (D) raised the inner exponent to the outer one, $3^2 = 9$ [misconception: the outer exponent applied to the exponent, not to the power]
7	$(-3)^2 \cdot (-3)^1$	-27	27 — the sign was dropped when the odd power was evaluated [7.NSO.1.1 clarification: rational-number bases include negatives]
8	$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$	$\frac{1}{32}$	1/6 — multiplied the exponents; 5/2 or 1/10 — treated the fraction as two numbers
9	Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?	No — $(10 + (-2))^3 = 8^3 = 512$	992 — distributed the exponent over the sum [B1G-M 7.NSO.1.1: the laws apply to products and quotients, never to sums]

Question Bank — variation and answers

Variation. Bank questions 1–5 vary one thing at a time inside the product law: two powers, then three, then evaluate-and-multiply against the law, then a negative base, then a case the student must judge (Jose, same base). Questions 6–8 move to the power of a product with the same three moves. The pattern breaks at question 13, where the exponent is the unknown, and at 15, where an equivalent expression has a different base ($4^3 = 2^6$). The Additional sheet mirrors the structure with different numbers and breaks at Leeland (different bases — the law does not apply).

The bank and its Additional sheet are posted, never assigned; they feed the quizzes. Answers below are the same values the keys print — both come from one source.

Question Bank

- $(3 \cdot 3 \cdot 3)(3 \cdot 3 \cdot 3 \cdot 3) = 3^7$
- $5 \cdot (5 \cdot 5 \cdot 5 \cdot 5)(5 \cdot 5 \cdot 5) = 5^8$
- $4 \cdot 64 = 256$**
- $(-8)^8 = 16,777,216$
- Yes — the bases are the same, so $9^3 \cdot 9^{12} = 9^{15}$.**
- $(4 \cdot 5)(4 \cdot 5) = (4 \cdot 4)(5 \cdot 5) = 4^2 \cdot 5^2$
- $(-35)^4 = 1,500,625$
- $3^2 \cdot (-6)^2 = 9 \cdot 36 = 324$

9. 3^{16}
10. $0.5^2 \cdot (-8)^2 = 0.25 \cdot 64 = 16$ — *power of a product*
11. (a) 285.6 (b) $\frac{24}{8} = 3$
12. (a) $n = 5$ (b) $n = 5$ (c) $n = 4$
13. A A
14. A, B, C, D A, B, C and D

Question Bank — Additional

1. $(4 \cdot 4)(4 \cdot 4 \cdot 4 \cdot 4 \cdot 4) = 4^7$
2. 7^6
3. $125 \cdot 25 = 3,125$
4. $(-6)^9 = -10,077,696$
5. *No — the bases are different (7 and 3), so the law does not apply. The value is $343 \cdot 2,187 = 750,141$, found by evaluating each power.*
6. $(7 \cdot 8)(7 \cdot 8) = (7 \cdot 7)(8 \cdot 8) = 7^2 \cdot 8^2$
7. $(-18)^4 = 104,976$
8. $2^3 \cdot (-9)^3 = 8 \cdot (-729) = -5,832$
9. $5^{16} = 152,587,890,625$
10. $2^{10} = 1,024$ — *power of a power*
11. (a) $n = 5$ (b) $n = 4$ (c) $n = 5$
12. A A
13. A, B, C, D A, B, C and D

What the audit against Math Nation found

- Math Nation 3.1 is sound page by page; all its worked values were recomputed (6^{12} , $(-8)^9$, $(-24)^5$, 6^8 , 10^{12} , 0.5^7 , 16).
- Additional Practice L1 #7 key: $(-18)^4$ printed 104,796; correct 104,976. Fixed on the Additional bank here.
- Additional Practice L1 #10 key stopped at 2^{10} without the value the column asked for; the bank item asks for and keys 1,024.
- The book's warm-up is a growth-mindset video with no mathematics in it; replaced by spaced retrieval.
- The book's explorations are all 'with your partner'; rebuilt as front-taught notes and boards (house rule 4).

What changed from the book

- 3.1.1 video warm-up → four retrieval questions.
- 3.1.2–3.1.5 partner explorations → Notes I–IV and two worked examples with Your Turns.
- 3.1.6 Your Turn and 3.1.7 Ticket Out the Door → the nine-question round; no exit ticket (ruling 10).

- Homework and Additional Practice → the two question banks, posted not assigned.